Hook-shaped Schur expansions are equivalent to uniformity of the descent-set distribution

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Editorial note. The counted statement is the author's labelled proposition 'prop:hook schur positive' (source lines 1334-1341); the proof environment begins on the next line after \end{prop} apart from one blank line, so the pairing is by adjacency.

Editorial note. The retained context is the author's own lead-in sentence introducing the criterion.

Editorial note. The retained preamble is the editorially prepared standalone preamble retained for this source, and the retained bibliography of the source is bundled because the retained proof cites a reference for the converse direction.

Editorial note. One external reference records the Schur-positivity criterion invoked by label inside the retained proof; it is a navigation pointer only.

Editorial scope. One original result of the pinned publisher article, counted once: the hook-shaped criterion for Schur-positivity (author label prop:hook schur positive).

Allowed standard prerequisites (editorial).

- Hook Schur function. s_{N-k,1^k} is the Schur function of the hook shape with k boxes below the first row.. It is the shape appearing in the claimed expansion.
- Descent set of a hook tableau. Every standard Young tableau of hook shape (N-k,1^k) has exactly k descents, and each k-subset of [N-1] arises from exactly one such tableau.. It is the bijective step in the forward direction at source line 1344.
- Schur-positivity criterion. A set is symmetric for D if and only if its generating function expands over standard Young tableaux with coefficients depending only on the shape.. It converts the displayed generating-function identity into the Schur expansion at source line 1349.

Source reference. Original theorem 2.5: sources/alco-351-matching-schur-2024/publisher-source.tex, source lines 368–368. Original criterion of Adin and Roichman invoked at source line 1349. It is a navigation pointer: the counted proposition's own proof is retained line for line and nothing is replaced by this reference. The reference number was inferred by counting the theorem-class environments of that section of the pinned source; it is not confirmed by the retained bibliography or by a label-number dump of the original, so it must be read as a navigation pointer only.

There is a simple well-known criterion for sets under which the generating function $\mathcal{Q}_D(\mathcal{A})$ is non-negatively spanned by the Schur functions of hook shape $s_{N-k,1^k}$:

Proposition 0.1 (folklore)**.** A set $\mathcal{A}$ with a statistic $D : \mathcal{A} \to 2^{[N-1]}$ has a generating function of the form

$$\mathcal{Q}_D(\mathcal{A}) = \sum_{k=0}^{N-1} c_k s_{N-k,1^k},\ c_k \in \mathbb{N}$$

if and only if for every $J \subseteq [N-1]$, $|\{a \in \mathcal{A} \mid D(a) = J\}|$ depends only on $|J|$. In that case, $c_k = |\{a \in \mathcal{A} \mid D(a) = [k]\}|$.

Proof. On one hand, assume that there exist $c_k \in \mathbb{N}$ such that $|\mathcal{A}(D = J)| = c_{|J|}$ for all $J \subseteq [N-1]$. Notice that every tableau of a hook shape $T \in \mathrm{SYT}(N-k, 1^k)$ has $|\mathrm{Des}(T)| = k$. Conversely, for every set $J \subseteq [N-1]$ with $|J| = k$ there exists a unique tableau $T \in \mathrm{SYT}(N-k, 1^k)$ such that $\mathrm{Des}(T) = J$. Therefore, we obtain the equation

$$\sum_{a \in \mathcal{A}} \boldsymbol{t}^{D(a)} = \sum_{k=0}^{N-1} c_k \sum_{T \in \mathrm{SYT}(N-k,1^k)} \boldsymbol{t}^{\mathrm{Des}(T)},$$

where $\boldsymbol{t}^J := \prod_{j \in J} t_j$ for $J \subseteq [N-1]$. By Theorem 2.5, this equation implies the statement.

The other direction of the proposition states that if $\mathcal{Q}_D(\mathcal{A}) = \sum_{k=0}^{N-1} c_k s_{N-k,1^k}$, then $|\{a \in \mathcal{A} \mid D(a) = J\}| = c_{|J|}$. Adin and Roichman [3, Corollary 8.1] established that a generating function uniquely determines the descent set distribution, which implies the statement. □

References

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